

UDL Beam

Example 1 - rivt Doc

user example

proj. 0001



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0.1-1 | Summary and Loads

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam using E-B theory [0.1.1]. It also serves as an annotated example of a single rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the [rivt user manual](#).

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command at the end of each rivt file (Doc-API rv.D). Published files are found in the _published folder.

0.1-2 | Load Combinations

Dead and live loads effects are taken from ASCE 7-05 [0.1.2]

Table 1: Load Effects (stored: t001-1.csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

0.1-3 | Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 p_sf	0.18 kPA	joists DL
D_2	2.10 p_sf	0.10 kPA	plywood DL
D_3	10.00 p_sf	0.48 kPA	partitions DL
D_4	3.00 k_ft	43.78 kN_m	fixed machinery DL
L_1	40.00 p_sf	1.92 kPA	ASCE7-05 LL
b_1	10.00 inch	254.00 mm	beam width



variable	value	[value]	description
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 k_si	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 p_si	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the rvsrsrc/data folder. The description is the table title, followed by the max column width.

Table 3: Beam Geometry (rvsrc/vals/beam1.csv)

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

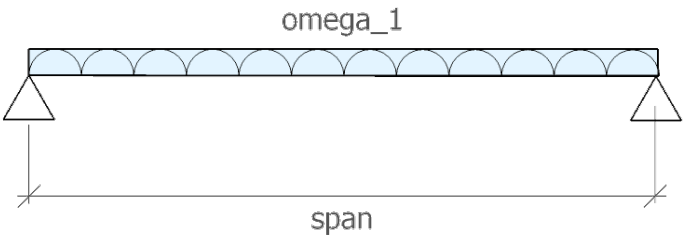


Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: Dead load [ASCE7-05 2.3.2]

$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$				
$dl_1 = 3.64 \text{ k_ft} \quad [dl_1] = 53.09 \text{ kN_m} \quad \text{ Dead load [ASCE7-05 2.3.2]}$				
spc_1	D_3	D_1	D_4	D_2
2.00 ft	10.00 p_sf	3.80 p_sf	3.00 k_ft	2.10 p_sf
beam spacing	partitions DL	joists DL	fixed machinery DL	plywood DL



Eq. 2: Live load [ASCE7-05 2.3.2]

$$ll_1 = 1.6 \cdot L_1 \cdot spc_1$$

$$ll_1 = 0.13 \text{ k_ft} \quad [ll_1] = 1.87 \text{ kN_m} \quad | \text{ Live load [ASCE7-05 2.3.2]}$$

L_1	spc_1
40.00 p_sf	2.00 ft
ASCE7-05 LL	beam spacing

Eq. 3: Total load [ASCE7-05 2.3.2]

$$w_1 = dl_1 + ll_1$$

$$w_1 = 3.77 \text{ k_ft} \quad [w_1] = 54.96 \text{ kN_m} \quad | \text{ Total load [ASCE7-05 2.3.2]}$$

dl_1	ll_1
3.64 k_ft	128.00 ft·p_sf
Dead load [ASCE7-05 2.3.2]	Live load [ASCE7-05 2.3.2]

0.1-4 | Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions (rvsrc/scripts/sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S (sectprop.py)

$$section_1 = rectsect(b_1, h_1)$$

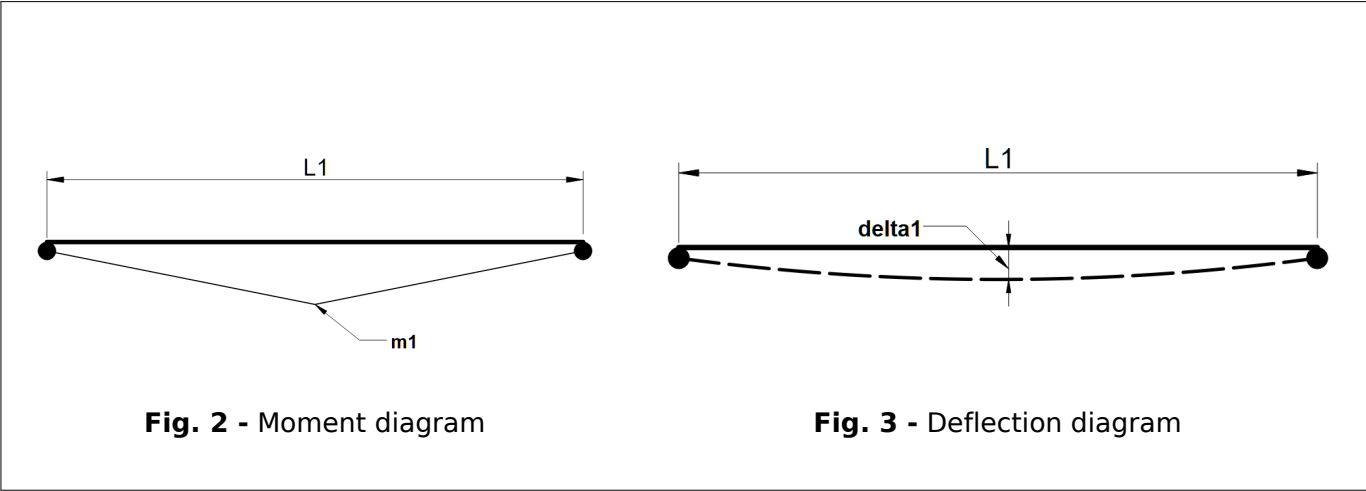
$$section_1 = 540.00 \text{ in}^3 \quad [section_1] = 8849.01 \text{ cm}^3 \quad | \text{ rectangle - S (sectprop.py)}$$



<u>b₁</u>	<u>h₁</u>
10.00 inch	18.00 inch
<u>beam width</u>	<u>beam depth</u>

Eq. 5: rectangle - I (sectprop.py)

inertia ₁ = rectinertia(b ₁ , h ₁)		
inertia ₁ = 4860.0 in4	[inertia ₁] = 202288.5 cm4	rectangle - I (sectprop.py)
<u>b₁</u>	<u>h₁</u>	
10.0 inch	18.0 inch	
<u>beam width</u>	<u>beam depth</u>	



Maximum bending stress formula

Eq.6:

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 7: Mid-span UDL moment



$$m_1 = \frac{\omega_1 \cdot \text{spn}_1^2}{8}$$

$m_1 = 120.52 \text{ ft-kip}$ $[m_1] = 163.40 \text{ mkN}$ | Mid-span UDL moment

ω_1	spn_1
<u>3.77 k_ft</u>	<u>16.00 ft</u>
Total load [ASCE7-05 2.3.2]	beam span
-	-

Eq. 8: Bending stress

$$fb_1 = \frac{m_1}{\text{section}_1}$$

$fb_1 = 2678.2 \text{ p_si}$ $[fb_1] = 18.5 \text{ MPA}$ | Bending stress

m_1	section_1
<u>120.5 ft²·k_ft</u>	<u>540.0 inch³</u>
Mid-span UDL moment	rectangle - S
-	(sectprop.py)

Eq.9: Stress ratio

$fb_1 < Fb_1$
 $[1] \text{ } fb_1$ $[2] \text{ } Fb_1$ ratio [1]/[2] check reference
 2.68 k_si 20.00 k_si 0.13 OK Stress ratio

Eq. 10: mid-span deflection (sectprop.py)

$$\delta_1 = \text{midspan}_\delta(\text{spn}_1, \omega_1, E_1, \text{inertia}_1)$$

$\delta_1 = 0.04 \text{ inch}$ $[\delta_1] = 1.00 \text{ mm}$ | mid-span deflection (sectprop.py)



<u>inertia₁</u>	<u>E₁</u>	<u>ω₁</u>	<u>spn₁</u>
4860.00 inch4	29000.00 k_si	3.77 k_ft	16.00 ft
rectangle - I (sectprop.py)	modulus of elasticity	Total load [ASCE7-05 2.3.2]	beam span -

[0.1.1]

“Euler-Bernoulli beam theory”, Wikipedia, Wikimedia Foundation.
[Online].https://en.wikipedia.org/wiki/Euler_Bernoulli_beam_theory. [Accessed: Jun. 15, 2026].

[0.1.2]

ASCE/SEI 7-05, Minimum Design Loads for Buildings and Other Structures, American Society of Civil Engineers, 2005.

